

ECON 100B Spring 2026
Midterm Exam ANSWERS

Rules and instructions:

1. You have 60 minutes to complete this in-class exam, worth 60 points in total.
You must stop working when time is called. If you continue working, you will be penalized 10 points, then an additional 10 points, and so on until you turn in your exam.
2. When time is called, **stop working**, stand up, and go give your exam to a proctor or place it in the receiving box. Do NOT give it to another student. Students who fail to turn in their exams promptly will receive the point penalty stated above.
3. This is an open-book, open-note exam, but **notes are limited to 1 page double-sided**. We suggest that you bring a **stand-alone calculator**. You might find a ruler useful.
4. PROHIBITED: Internet, texting, all communications. No laptops, smartphones, screens.
5. Your work on this exam must be your own. Violations earn a zero on the exam and a referral to the Center for Student Conduct. Sign the Berkeley Honor Code below:

"As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others."

(your signature)

6. Write legibly. If we can't read your answer, we can't award credit. Confine your answers to the space provided (inside the box). Don't fill the space with words. Less is more.
7. Restroom breaks are allowed. Give your phone to a proctor and retrieve it afterward.
8. **If you see something, say something to a proctor.** Pretend to go to the restroom and tell a proctor briefly what's going on. If you prefer, write something on your exam instead. If you are the victim of academic dishonesty by another student, take countermeasures.

(1 point) Write your name and student ID at the top of **each page** of the exam.

(1 point) If you are in Pimentel Auditorium, record your **row letter and seat number**.

(1 point) What is $2 + 2$? (George Orwell fans earn extra praise, but all students earn this point.)

1. (15 points total) Short questions, short answers, Get Shorty (1995)¹

- (a) (3 points) If you were the leader of a country, and you had the choice between more capital formation and more idea production, which one would you choose and why?

Idea production wins, hands-down. Capital formation is important, but idea production is a stronger source of long-term growth in income per person. Ideas are nonrival, but capital isn't. The Romer model reveals that TFP growth can derive from idea production.

- (b) (3 points) For a profit-maximizing firm, what do we typically call the marginal product of labor? What do we typically call the marginal product of capital?

The wage, or earnings, is the usual name for the marginal product of labor. The **interest rate** is the usual name for the marginal product of capital. A more obscure term but also correct is **rents**. A generic term would be returns to capital.

- (c) (3 points) True or false, and briefly discuss: Holding everything else constant, a country with more population growth will be a country with higher living standards (meaning income per person).

False. The Solow model shows us that the population growth rate and income per person typically move in opposite directions. Given a particular saving rate, a faster rate of population growth will reduce steady-state capital per person and income per person.

- (d) (3 points) In India, the annual growth rate of GDP is about 8%, and the annual growth rate of population is about 1%. What is the growth rate of Indian GDP per person? Briefly show your work. Does this number seem low or high compared to growth rates in advanced economies like the U.S., Europe, and Japan?

The growth rate of a ratio is the difference of the growth rates: $8 - 1 = 7$ percent. This number is **high** relative to comparable statistics in the U.S., Europe, and Japan, where income per person usually grows at 2% or less.

- (e) (3 points) Suppose a country had weak societal institutions, meaning that its government could not protect its citizens against theft, and that government sometimes stole resources from its citizens and threatened to imprison them if they objected. What would citizens of this country likely do if they could? What would they likely not do?

Full-credit answers can range widely here. Citizens of a country like this would be likely to migrate away if they could, or to invest their resources elsewhere rather than domestically. They seem likely to delay childbearing if possible. They likely would not invest resources in the local economy. They might not save and invest in anything that could be stolen.

¹ Note: Here and throughout the exam, take any political views that you like, or none at all. Scores reward completeness of answers only and will not penalize any political viewpoint or the lack of one.

2. (22 points total) Saving rates and other things across Solow economies

Recall the Solow model expressed in ratios, with capital per worker $k = K/L$ on the horizontal axis. When output is Cobb-Douglas with capital share equal to $1/3$,

$$\text{capital per worker} \quad k = K/L \quad (2.1)$$

$$\text{output per worker} \quad y = A k^{1/3} \quad (2.2)$$

$$\text{saving per worker} \quad s y = s A k^{1/3} \quad (2.3)$$

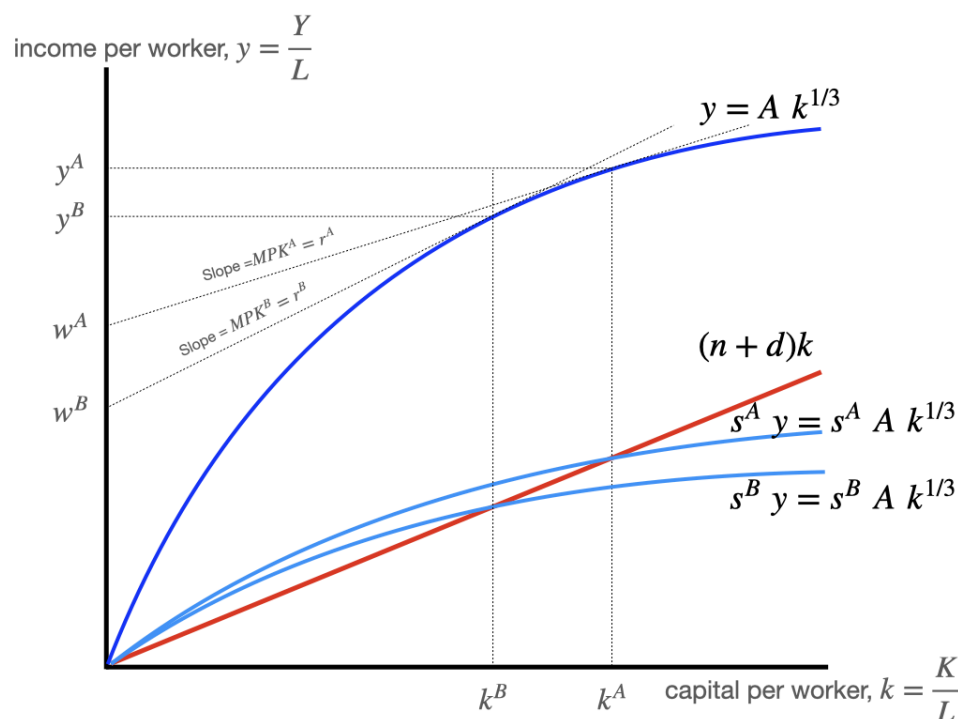
$$\text{depreciation per worker} \quad (n + d)k \quad (2.4)$$

$$\text{marginal product of capital} \quad MPK = \frac{1}{3} A \left(\frac{L}{K} \right)^{2/3} = \frac{1}{3} A k^{-2/3} \quad (2.5)$$

$$\text{marginal product of labor} \quad MPL = \frac{2}{3} A \left(\frac{K}{L} \right)^{1/3} = \frac{2}{3} A k^{1/3} \quad (2.6)$$

Consider two Solow economies in which $n = 0.01$ and $d = 0.04$ and the level of total factor productivity A is the same, but the saving rate in Solow economy A is $s^A = 0.20$, while in economy B it is $s^B = 0.15$. Assume these two economies have the same level of population.

- (a) (4 points) Depict both Solow economies on the same axes below. Show output per worker, saving per worker, and depreciation per worker for each country. Some or all of these may be the same for both countries. Show the steady state(s). Discuss what you have drawn. (Explicitly showing the wage and MPK is optional but might be helpful.)



When the population growth rate, depreciation rate, the level of TFP, and the capital share are all the same, then the output curve and depreciation curve are the same. But the saving curve is higher for country A, and so are its steady-state levels of capital and output per worker.

- (b) [2 points] In which country is the **wage** higher, or is the wage the same in both countries? Briefly explain, referencing math or the graph or both.

The wage is higher in **country A**, where the saving rate and steady-state capital per worker are also higher. Equation (2.6) shows that this the $MPL = w$ will be higher in the country with the higher k , and we can also see it in the graph I have drawn, where the wage is the level on the vertical axis where the tangent line, which has slope equal to $MPK = r$, hits.

- (c) [2 points] In which country is the **interest rate** higher, or is the interest rate the same in both countries? Briefly explain, referencing math or the graph or both.

The interest rate is higher in **country B**, where the saving rate and steady-state capital per worker are lower. This is shown by equation (2.5), which reveals that a higher level of k will produce a lower level of MPK , and thus a lower k in country B will produce a higher MPK . In the graph, the tangent line, which has slope equal to MPK , is steeper for country B.

Factor inputs like labor and capital probably seek higher returns, meaning that if labor and capital are free to move across borders, each one will try to move to where its return is highest.

- (d) [2 points] Under the circumstances described above, where would **labor** seek to migrate? Into country A, into country B, or would labor seek to remain where it is? Take a stand and briefly explain.

Labor would like to move **from country B into country A** if possible. Country A has a higher wage because its capital-labor ratio is higher, and labor would prefer to reallocate toward country A.

- (e) [2 points] Under the circumstances described above, where would **capital** seek to move? Into country A, into country B, or would capital seek to remain where it is? Take a stand and briefly explain.

Capital would prefer to move **from country A into country B** if it can. Country B has a higher interest rate because its capital-labor ratio is lower, so capital would prefer to reallocate toward country B.

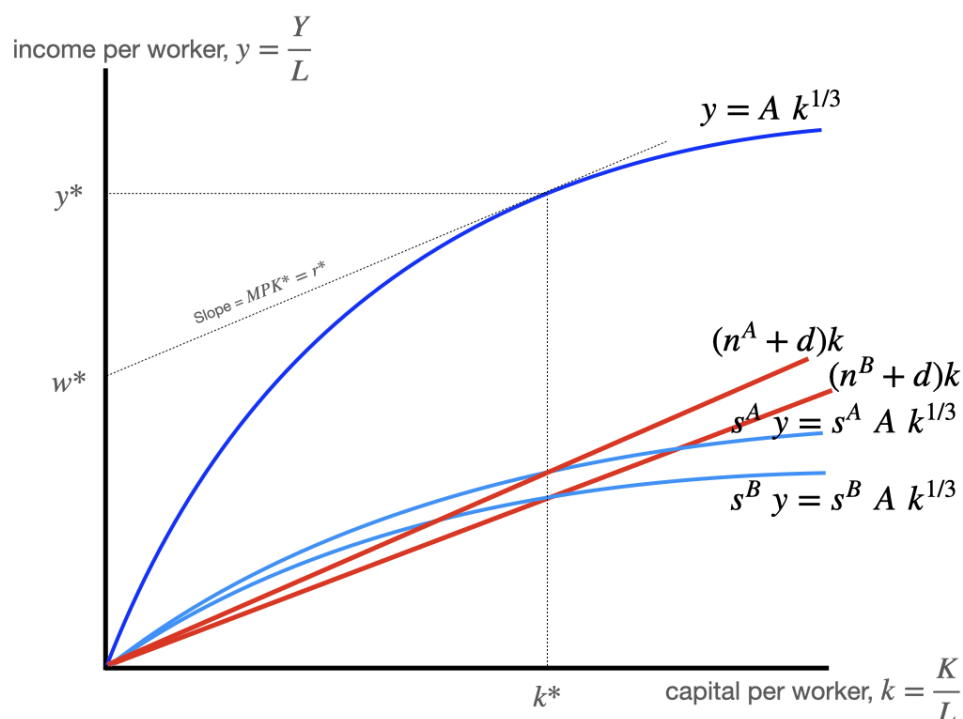
- (f) [2 points] Suppose labor was restricted from moving across international borders but capital could move costlessly. Qualitatively speaking, how would saving per worker shift between country A and country B, if any, and in which direction? Make a reference to the graphs that you drew.

Full points are earned by: Savers in country A would shift their excess saving into country B, raising the saving curve in country B and reducing it in country A in what we drew. (If the two countries had populations of the same size and the same level of TFP, savers in country A would shift half their excess saving into country B, $s^A - s^B$, to equalize saving per worker across the two countries.) This would also equalize the interest rates and the wages.

- (g) [4 points] Recall the scenario in part (f). Suppose saving could be exported as manufactured capital goods. Which country would be running a trade deficit, meaning that it imports goods from a different country? Discuss.

When savers in country A shift their excess saving into country B, **country B runs a trade deficit** by importing the capital goods produced in country A. The country with excess saving runs a trade surplus, exporting to the country with lower domestic saving. (The mechanism isn't demand for goods; it's equalizing factor prices, or if capital is mobile, the interest rate.)

- (h) (4 points) Suppose capital was restricted from moving across borders but labor could move both costlessly *and continuously, not just once*. Draw a new Solow graph that shows both economies on the same axes below, but illustrate the effects of continuous labor migration that fully offsets the effects of the original differences in the saving rates between countries A and B on wages. Show output per worker, saving per worker, and depreciation per worker for each country. Some or all of these may be the same for both countries. Show the steady state(s). Discuss what you have drawn. (Explicitly showing the wage and MPK is optional but might be helpful.)



If labor can reallocate continuously to seek higher returns, then labor from country B will shift continuously toward country A, decreasing country B's population growth rate to n^B and increasing country A's population growth rate to n^A as shown. The red depreciation line pivots clockwise for country B, raising its capital per worker up to k^* , and the it pivots counter-clockwise for country A, reducing its capital per worker down to k^* . Both countries still have the same level of TFP, and so their output per worker, interest rate, and wage are all the same if the population continuously shifts so that k^* is the steady-state capital per worker in both countries.

3. [20 points total] Brain drain and brawn drain in a drawn Romer economy

Consider two “old school” Romer economies A and B, with production of knowledge A_t^i by scientists L_{at}^i and of food Y_t^i by farmers L_{yt}^i . Quantities in the economies are denoted with a superscript $i = A, B$ so that output in economy A is written Y_t^A . The equations are familiar:

$$\Delta A_t^i = z^i A_t^i L_{at}^i \quad (3.1)$$

$$\bar{N}^i = L_{at}^i + L_{yt}^i \quad (3.3)$$

$$Y_t^i = A_t^i L_{yt}^i \quad (3.2)$$

$$L_{at}^i = \ell^i \bar{N}^i \quad (3.4)$$

Assume nothing is shared across the two economies. Recall the solution, plus a helpful elaboration about what income per person is, in equation (3.8) below:

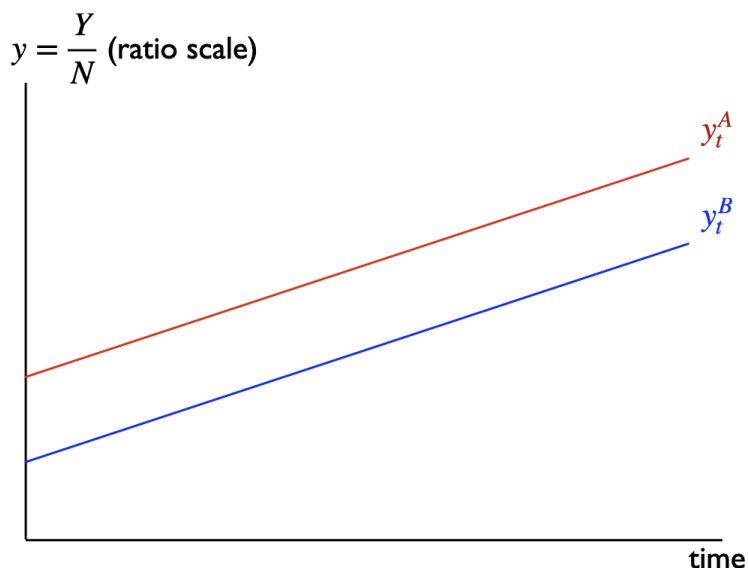
$$g[A^i] = z^i \ell^i \bar{N}^i \quad (3.5)$$

$$A_t^i = A_0^i (1 + g[A^i])^t \quad (3.6)$$

$$y_t^i = A_0^i (1 - \ell^i) (1 + g[A^i])^t \quad (3.7)$$

$$y_t^i = \frac{Y_t^i}{L_{at}^i + L_{yt}^i} = \frac{A_t^i L_{yt}^i}{L_{at}^i + L_{yt}^i} \quad (3.8)$$

- (a) [4 points] Suppose that the only difference between these two Romer economies is that economy A started with a higher level of ideas: $A_0^A > A_0^B$. All other parameters are the same. Graph y_t^A and y_t^B on the same axes below and briefly explain what you’ve shown. Discuss levels and growth rates. Be precise and succinct.



When economy A starts with a higher level of ideas but all other parameters are the same, the **growth rates** of the two economies **are the same**, so the trajectories of y are parallel on a ratio scale, but there is a permanent **difference in levels** between them.

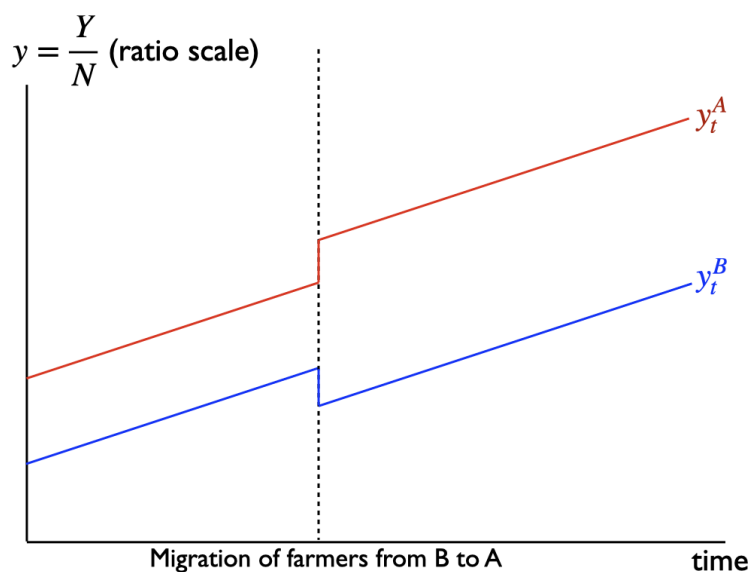
- (b) [2 points] Consider a **farm worker** in country B. If this worker had the option to migrate to country A, would the worker want to do so? Assume workers have access to local technology and average incomes. Take a stand yes or no, and make a brief argument.

Yes, a farm worker in country B would want to migrate to country A if it were possible. Income per person $y = Y/N$ is higher in country A than it is in country B, and that means that the farmer's income would rise after migrating. Intuitively, the level of ideas is higher in country A, and that means income per person and also the farm wage are both higher in country A.

- (c) [2 points] Consider a **scientist** in country B. If this scientist had the option to migrate to country A, would the scientist want to do so? Assume workers have access to local technology and average incomes. Take a stand yes or no, and make a brief argument.

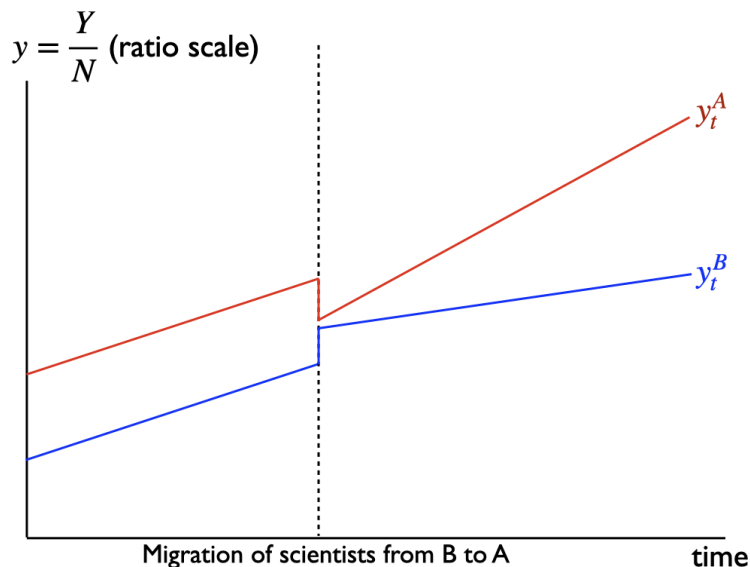
Yes, a scientist in country B would also want to migrate to country A. The logic is the same as for the farm worker in part (a). Income per person is higher in country A, and the scientist wage is also higher in country A.

- (d) [4 points] Suppose that a large number of **farmers** migrate from country B to country A and remain farmers. Depict what would happen, if anything, to income per person in both countries. Briefly explain. Discuss levels and growth rates. Be precise and succinct. *Hints: Pay special attention to numerators and denominators. Look at equation (3.8).*



When a large number of **farmers** migrate from country B to country A, their departure **has no effect on the growth rate** of either. The growth rate depends on the productivity of scientists and the number of resident scientists, neither of which has changed. But their leaving **reduces output per person in country B and increases it in country A**, because output is produced by ideas and farmers. When farmers leave country B, it is left with fewer farmers and less output but the same number of scientists. Output per remaining resident falls. The same logic shows that country A gains farmers and output, so output per resident rises.

- (e) [4 points] Now suppose that a large number of **scientists** migrate from country B to country A and remain scientists. Depict what would happen, if anything, to income per person in both countries. Briefly explain. Discuss levels and growth rates. Be precise and succinct. *Hints: Pay special attention to numerators and denominators. Look at equation (3.8).*



When a large number of **scientists** migrate from country B to country A, their departure **reduces the growth rate in country B** and **increases the growth rate in country A**. The number of resident scientists rises in A and falls in B. When scientists leave country B, their departure also causes an upward jump in output per person, because output depends on the current stock of ideas and the number of farmers. With fewer resident scientists in country B, output per person actually rises in the short run. But over the long run, the departure of scientists reduces the growth rate in country B. (The shapes of the two lines can be different from what is shown. The blue line could jump up above the red line if the relocation is large, and it could also become flat but can't become steeper. The red line must become steeper.

- (f) [4 points] Discuss what your results imply about the macroeconomic effects of immigration in the Romer model. Does the average resident (who enjoys income per person) of immigrant receiving countries like country A always benefit from immigration? Does it depend on the type of worker who migrates? Does it change over time?

The picture is **somewhat mixed**, without a single answer about macroeconomic effects of immigration. But **in the long run, residents in the receiving country always benefit** from immigration of either type of worker in this model. Country A is better off immediately and permanently with more farmers, and it is also better off in the long run with more scientists. But **in the short run, residents of country A actually suffer** a reduction in income per person **when scientists immigrate**. Their farm production does not increase immediately, and thus income per person falls in the short run, because the same number of farmers are supporting more scientists. Income per person eventually increases above the original balanced growth path in country A. Residents of **country B** actually see a **short-run benefit** when scientists leave, but in the long run they suffer. They always suffer when farmers leave.